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PI Name:	Minna, John D. M.D.		
Project Title:	NSCOR: Risks Estimates and Mechanisms of Lung Cancer Pathogenesis after Space Radiation		
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Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Elements:	(1) **SR**:Space Radiation		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-8593	**Congressional District:**	30
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Dr. Min Chen, Biostatistitian, secured DOD funding for his own project and was required to relinquish his effort on this project effective February 2013. Dr. David Chen, Project 3 Lead, was on sabbatical beginning January 2013. He remained in close contact with project investigators and returned from sabbatical effective September 1, 2013.		
COI Name (Institution):	Asaithamby, Aroumougame (The University of Texas Southwestern Medical Center) Chen, David (The University of Texas Southwestern Medical Center) Gazdar, Adi M.D. (The University of Texas Southwestern Medical Center) Girard, Luc Ph.D. (The University of Texas Southwestern Medical Center) Shay, Jerry Ph.D. (The University of Texas Southwestern Medical Center) Story, Michael Ph.D. (The University of Texas Southwestern Medical Center) Xie, Jin (The University of Texas Southwestern Medical Center) Xie, Yang Ph.D. (The University of Texas Southwestern Medical Center)		
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Task Description:

The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes are being compared in these models to that of gamma-irradiation. These dose risk assessments are designed to measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They also measure effects on development of cancer stem-like cells (cancer initiating cells). 3D organotypic culture and animal models also allow measurement and risk estimation in both tissue surrogates and in vivo lung tissues. These individual risk estimates can then be combined into a model for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORs and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have completed many studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified and that specific mRNA expression profiles for HZE irradiation exist. In addition, we have found that transgenic mouse models of lung cancer have increased lung cancer risk particularly after fractionated high LET exposure allowing us to study the early onset and progression of lung cancer in the whole animal in real time. The NSCOR has 4 Projects:
1. HZE Particle Exposure and the Risk for Human Lung Carcinogenesis;

2. Mouse Models of Lung Cancer after HZE Particle Irradiation;

3. Organotypic (3D) Human Models of Genomic Instability for HZE Particle Induced Lung Carcinogenesis;

4. Integrating Biomarkers of Lung Cancer Pathogenesis for Risk Assessment after HZE Particle Irradiation.

These projects are supported by 2 Cores: Administrative Core and the Database, Biostatistics and Bioinformatics Core. The HZE particle irradiation is done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, and a Center for the NCI Early Detection Research Network (EDRN) program, providing great synergism and additional resources available for the successful completion of this proposal.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused mostly by smoking exposure, a significant number of lung cancers arise in life time never smokers and thus is a sizable health problem in the USA. Indeed lung cancer in never smokers is the 7th leading cause of world-wide cancer related deaths. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high-LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. In addition, in the past decade in the USA, Europe, and Asia there has been a resurgent interest in radiographic screening for lung cancer, particularly using serial computed tomography (CT) scans. In fact, the USA NCI has completed a large (50,000 person) National Lung Cancer Screening Trial (NLST) that randomized persons to have CT scans or not for the early detection of lung cancer. This trial has found a significant survival benefit for persons receiving CT scan screening. This raises a variety of questions concerning the use of this approach because of its cost and substantial population based radiation exposure. Greater than 25% of persons screened (current or former smokers) will have some lesions requiring further CT scan follow-up, when biopsies are performed only 1% of the overall group will have histologic evidence of lung cancer. In addition, many of the persons will require several CT scans in follow-up to determine if the lesion is growing to warrant a biopsy. Thus, there is considerable interest in knowing the contribution of CT scan doses of gamma radiation to the pathogenesis of lung cancer as well as other environmental exposures. This is particularly true since this radiation is being administered to individuals (current and former smokers) who are at highest risk of having preneoplastic lesions leading to lung cancer. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 50 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as expression of oncogenic K-RASv12, knockdown of expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high-LET and low dose gamma irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high-LET radiation in space in order to reduce the uncertainties in prediction of risk of astronauts developing lung cancer as part of long-term missions in space. Thus, the results will also have major impact and benefit to life on Earth that includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through CT scan screening studies, and environmental exposures, accidentally or through terrorism. A part of this will be to understand the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional (3D) cultures, as well as growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes.
Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions: "We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and

molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models" ; "We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate" ; "We are developing tissue specific risk models using human 2D and 3D culture as well as animal models for lung cancer" ; "We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated". In summary, we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. Thus, our assembled working team is synergistically contributing more to assessing cancer risk associated with space radiation than would be accomplished by study of the individual components in isolation.

Our lung cancer NSCOR is addressing the following specific critical path roadmap and gap questions:

• We are improving the understanding of space radiation in lung cancer initiation, promotion and progression spanning cell and molecular biology at the cell, tissue and system biology level that will lead to significant reduction in the uncertainties in risk projection models

• We are investigating the increased risk from space radiation as a function of age of exposure, age, latency, gender, tissue, radiation quality and dose rate

• We are developing tissue specific risk models using human 3D culture and animal models for lung cancer

• We are investigating how aberrant DNA damage processing, genomic instability, epigenetic effects including methylation, persistent oxidative damage, altered senescence, and non-targeted effects contribute to irradiation-associated lung cancer using a variety of surrogate endpoints that are being validated

Overall we are conducting research to investigate how biological endpoint and systems biology approaches can be used to integrate research on the cellular and molecular as well as tissue level of radiation damage that will lead to significant reductions in the uncertainties in risk projection models of lung cancer. As will be described in this report, we are confident and will provide compelling evidence that the sum of the efforts of our assembled team are significantly contributing more to assessing cancer risk associated with space radiation compared to the individual components.

Task Progress:

Project 1: Human lung epithelial cell models of HZE particle and gamma radiation effect on cancer progression. This project focuses on studies of the HBEC (human bronchial epithelial cell) system after HZE and gamma radiation combining biologic and molecular studies led by Dr. Michael Story and is extremely well aligned with the goals of the NASA mission. Strengths include the fact that the project has developed data for a variety of HZE particles and gamma radiation doses and fraction for survival, and transformation to allow soft agar colony growth after irradiation. Studies on "preinitiated" HBECs with selective oncogene changes were also studied and found to be new "biosensors" of HZE particle induced transformation. Using genome wide molecular analyses it has identified strong candidates for radiation carcinogenesis-related gene expression signatures. Future work focuses on very low dose, radiation quality and dose-rate effects which will provide important and high priority data.
Project 2: Mouse transgenic models of lung cancer progression after HZE particle and gamma irradiation. This interesting and important project is led by Dr. Jerry Shay, generating data that has already paid dividends which can be extrapolated to human lung cancer outcomes. The evidence that Fe-, Si-, and Protons when given in fractionated (protracted) doses increase the incidence of more lethal invasive lung cancer in the LA-1 K-ras mouse model is convincing. In addition, the solar particle event (SPE) simulation performed has indicated an increase in the incidence, promotion, and progression of lung cancer. Drawing a molecular connection between very early events in the mouse lung after HZE particle radiation with similar mRNA expression patterns in human lung cancers that have poor prognosis is a novel and interesting discovery and provides a mouse-human connection. Currently lower dose experiments are in progress and molecular underpinning of the mechanisms will be completed.

Project 3: DNA repair after HZE particle radiation in 2D and 3D HBEC cultures. The investigators (Drs. David Chen and Asaithamby Aroumougame) have developed several novel approaches for studying the effects of DNA double strand break (DSB) induction following HZE particle irradiation in human bronchial epithelial cells (HBECs). To determine the molecular bases for the cells' inability to repair complex DNA double strand breaks (DSBs) in 3 dimensional tissue-like culture (3D culture), they performed a global gene expression analysis using microarrays, as well as screened for chromosomal aberrations in metaphase cells derived from monolayer and 3D-organotypic cultures. They find that repair of the complex DSBs induced by high-LET iron particles, but not the isolated DSBs induced by gamma-rays, are substantially compromised by 3D organotypic culture. This appears to be related to decreases in a protein, RAD51, that is critical for homologous recombination repair of DNA double strand breaks. Thus, HZE particle radiation will have a greater effect when delivered in tissue like conditions. This system will also allow integration and study of microenvironmental changes into the 3D in vitro system.

Project 4: Using systems biology and bioinformatics to integrate molecular data following HZE particle and gamma radiation. This project (led by Dr. Yang Xie) also provides support as a Biostatistical and Bioinformatics Core (see below), to facilitate data compilation and to exploit bio-informatics to validate mutations, with regard to potential cancer predisposition. The goal of this project is to use computational biology approaches to integrate lung cancer risks and molecular biomarkers from the other projects to develop risk assessment models for increases in fatal lung cancers from exposure to space radiation. Developing new tools for systems biology analyses, this project has identified a small number of master regulating "hub genes" which predict expression changes in many other genes after various types of radiation. These "hub genes" will be the focus of detailed functional and mechanistic studies and integration of human and mouse findings going forward.

Administrative Core: (Minna, Shay, Davis) This core organized many meetings with internal and external reviewers and coordinated all travel plans to the NSRL and with visits to other NSCORs.

Bioinformatics, Biostatistics, Database Core: (Y.Xie, L. Girard) Aim 1: To provide valid statistical designs of experiments and conduct innovative statistical modeling, simulations, and data analyses. Aim 2: Data integration and quality control. Aim 3: To expand the current UTSW-NSCOR database to an ontology-driven and internet accessible database to allow data sharing and integrative analyses between NASA investigators.

	Overall Summary of Progress: • HZE Particle Radiation Induces Malignant Transformation: From human lung epithelial cell culture (HBEC) and in vivo mouse studies we have quantitated how protracted/fractionated HZE particle irradiation increases lung cancer progression toward malignancy with several ion types. • HZE Particle Radiation Activates Signaling Pathways that Promote Malignancy: From studies of HBEC and mouse models of lung cancer, we have determined quantitatively that altered signaling for ROS metabolism, inflammatory responses and DNA damage signaling can distinguish terrestrial from space irradiation and that similar pathways are activated in the most aggressive human lung cancers. • Importance of Rad51-dependent Homologous Recombination following HZE Particle Radiation: We have determined that DNA lesions induced by HZE particle are preferentially repaired by Rad51-dependent homologous recombination and that this repair is actually down-regulated under conditions similar to those found in tissues. • Identification of Key “Hub Genes” Following Exposure to HZE Particle Radiation: We have developed computational biology tools and approaches that allow us to identify a small number of genes which act as master regulators (“hub genes”) of gene expression programs in human lung cells following g and HZE particle radiation providing a roadmap for detailed functional and mechanistic studies. • Establishment of Database, Datasets, and Biospecimens to Facilitate NASA Collaborations: We have developed a robust database for sharing among our NASA team and others in the NSCOR lung consortium to permit sharing of biospecimens and integrative molecular analyses - using Super-SPACE model to predict space risk (561 human HBEC microarrays and 144 mouse microarrays of g and HZE irradiations).
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